

Airflow and Thermal Model of the CalSol Excalibur Battery Box

Reduced-order analysis from the Main Battery CAD

September 30, 2026

Abstract

A reduced-order airflow and thermal model of the CalSol Excalibur battery enclosure, built on geometry measured directly from the `Main Battery STEP` export rather than reconstructed. The pack is a staggered bank of 408 LG INR18650 MJ1 cells cooled by 80 mm fans mounted in two rectangular cutouts on opposite ends of the box.

Three results stand out. First, the cell bank accounts for 97.7% of the system resistance and the fans sit at 99% of their dead-head pressure, so *adding fans in parallel is nearly useless*: going from three to twelve buys 0.10°C for 227 W. Second, staging fans *in series* — pushing at one cutout and pulling at the other — raises the flow from 93.2 CFM to 131.6 CFM and drops the hottest cell from 47.02°C to 45.12°C at identical power, essentially reaching the 45°C design goal with no change to the hardware. Third, a configuration of one fan on each face is both cooler and cheaper than the current three-on-one-face arrangement (46.18°C at 50.4 W against 47.02°C at 75.6 W).

A transient lumped-node model with seeded manufacturing scatter is included, along with an explicit statement of where the model’s assumptions are weakest.

1 Geometry

All geometry is read from `fullbatterybox/Main Battery.step` (57 MB, 3374 solids) through the OpenCASCADE XCAF reader, which preserves part names and colours from the assembly tree. Bodies are classified by CAD part name; the 408 cells are identified by name and confirmed by shape.

Table 1: Pack geometry, measured from the STEP. The cell diameter is taken from the *cylindrical face* rather than the bounding box: the box reads 19.92 mm because it includes the terminal button, and that difference moves the inter-cell gap from 0.58 mm to 2.10 mm — a factor of 3.6 in the quantity that sets the pressure drop.

Quantity	Symbol	Value
Cells	N	408
Arrangement	—	$17 \times 12 \times 2$, staggered
Cell diameter	D	18.40 mm
Cell length	L	65.00 mm
Transverse pitch	S_T	20.50 mm
Longitudinal pitch	S_L	25.20 mm
Diagonal pitch	S_D	27.20 mm
Minimum gap	$S_T - D$	2.10 mm
Free-area ratio	$A_{\min}/A_{\text{face}}$	0.1024
Bank frontal face	—	246.0 mm $\times$ 130.0 mm
Total wetted area	A_{tot}	1.533 m ²

1.1 The openings

The enclosure is pierced by two large rectangular cutouts on opposite ends of the z axis, 590 mm apart, with the cell block centred between them. **Battery Box.step** is the only file in the set exported with true B-rep solids, so the cutouts could be measured as inner wires on planar faces rather than estimated.

Table 2: Measured openings. Each fits exactly three 80 mm fan frames across its width, independently confirming the design review’s “3 across the pack”. The rear cutout is partly blocked by an internal truss.

Opening	CAD part	Size	Porosity	Free area
Blue, front	Front Plate-1	251.3 × 121.0 mm	1.000	304.1 cm ²
Green, rear	Rear Plate-1	244.0 × 124.0 mm	0.886	268.1 cm ²
Front duct	Front Duct-1	247.3 × 117.0 mm	1.000	289.3 cm ²

The blue cutout is 251.3 mm wide against a bank face of 246.0 mm: a 2 % match. The openings were sized to the bank, which fixes the flow axis as z — air crosses the 17 rows.

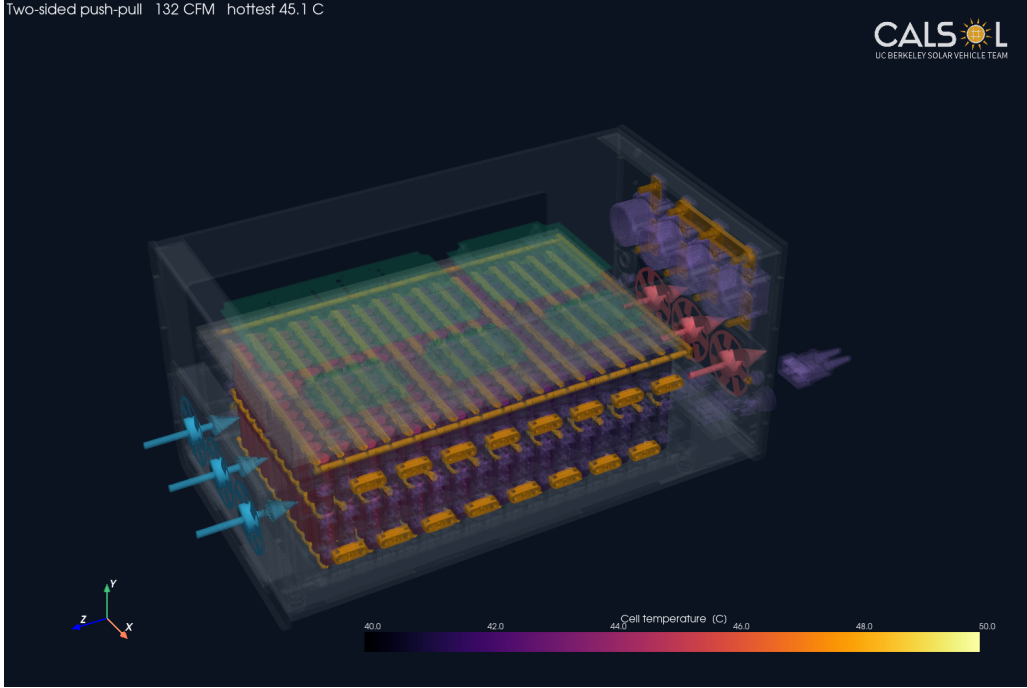


Figure 1: The assembly as modelled, with the enclosure drawn translucent. Blue glyphs are intake fans at the **Front Plate** cutout, red are exhaust at the **Rear Plate**; cells are coloured by computed temperature. The purple bodies at the far end are the four Gigavac GX11 contactors.

2 Governing equations

2.1 Pressure drop and the operating point

The bank drop follows the Žukauskas correlation for staggered tube banks,

$$\Delta p_{\text{bank}} = N_L \chi f \frac{\rho V_{\text{max}}^2}{2}, \quad (1)$$

where V_{max} is the velocity in the minimum free area, obtained from continuity,

$$V_{\text{max}} = V_{\infty} \frac{S_T}{\min(S_T - D, 2(S_D - D))}. \quad (2)$$

For this pack the transverse plane binds and $V_{\max} = 9.76 V_{\infty}$. The operating point is the flow at which the fan characteristic meets the system resistance,

$$\Delta p_{\text{fan}}(Q) = \Delta p_{\text{sys}}(Q) = \Delta p_{\text{bank}}(Q) + \Delta p_{\text{duct}}(Q), \quad (3)$$

solved by Brent's method with the air state iterated to consistency, since density depends on how much the air has warmed and that depends on the flow.

2.2 Fan composition: the central result

Fans side by side on one face are in *parallel* — equal pressure, flows add:

$$Q(\Delta p) = \sum_i Q_i(\Delta p). \quad (4)$$

Fans on opposite faces of the same duct are in *series* — equal flow, pressures add:

$$\boxed{\sum_i \Delta p_i(Q) = \Delta p_{\text{sys}}(Q)}. \quad (5)$$

Which law applies is the difference between a useful fan and a wasted one, for the reason developed in §4.1.

Continuity over the box closes the system:

$$\sum_{i \in \text{inlets}} Q_i = \sum_{j \in \text{outlets}} Q_j. \quad (6)$$

Equation (6) has no solution for $Q > 0$ if every opening carries a fan blowing inward: the box simply pressurises until one face reverses. The implementation raises an error in that case rather than returning a plausible number.

2.3 Heat transfer

The convective coefficient comes from the Žukauskas correlation,

$$\text{Nu} = C_1 C_2 \text{Re}_{\max}^m \text{Pr}^{0.36} \left(\frac{\text{Pr}}{\text{Pr}_s} \right)^{1/4}, \quad h = \frac{\text{Nu} k}{D}, \quad (7)$$

with C_1 a function of S_T/S_L and C_2 the shallow-bank correction. Air is marched row by row, its properties re-evaluated at each row's local temperature:

$$\dot{m} c_p \Delta T_{\text{air},r} = \sum_{i \in r} \dot{Q}_i, \quad T_{\text{cell},i} = T_{\text{air},r(i)} + \frac{\dot{Q}_i}{h A_i}. \quad (8)$$

2.4 Transient model

For time-dependent behaviour each cell becomes a lumped node of capacity $C_i = m c_p$, coupled to its neighbours by conduction and to the local air by convection:

$$C_i \frac{dT_i}{dt} = \dot{q}_i - h_i A_i (T_i - T_{\text{air},r(i)}) + \sum_{j \in \mathcal{N}(i)} G_{ij} (T_j - T_i). \quad (9)$$

Manufacturing spread enters physically, through the cell properties rather than as noise added to the result:

$$\dot{q}_i = I_{\text{cell}}^2 R_i, \quad R_i = R_{\text{int}} (1 + \epsilon_i), \quad \epsilon_i \sim \mathcal{N}(0, \sigma_R^2), \quad (10)$$

drawn once from a seeded generator so a run is reproducible.

Equation (9) is integrated with explicit Euler. Stability requires

$$\Delta t < \min_i \frac{C_i}{h_i A_i + \sum_j G_{ij}} = 29.3 \text{ s}, \quad (11)$$

so the chosen $\Delta t = 0.5 \text{ s}$ carries a $59\times$ margin. The system is not stiff — convection dominates conduction by roughly an order of magnitude — so an implicit scheme would cost a dense solve per step and buy nothing. The air is treated as quasi-steady: its transit time through the bank is 0.31 s against a cell time constant of 39.9 s , a separation of $128\times$.

3 Operating conditions

Table 3: Inputs, each tagged with its provenance. Nothing here is invented; where a value could not be sourced it is marked **ASSUMED** and surfaced in the tool.

Parameter	Symbol	Value	Source
Pack configuration	—	34S12P	REVIEW
Cell internal resistance	R_{int}	28 m Ω	REVIEW
Pack voltage	V	85 V	REVIEW
Worst-case draw	P	5200 W	REVIEW
Pack current	I	61.18 A	derived
Cell current	I_{cell}	5.10 A	derived
Pack heat load	$\dot{Q}$	296.9 W	derived
Per-cell heat	$\dot{q}$	0.728 W	derived
Inlet air temperature	T_{∞}	40 °C	REVIEW
Design goal	—	45 °C	REVIEW
Absolute limit	—	60 °C	REVIEW
Fan	—	Delta GFB0812ES-E, 25.2 W	DATASHEET
Cell mass	m	48 g	DATASHEET
Cell specific heat	c_p	1050 J kg ⁻¹ K ⁻¹	LITERATURE
Radial conductivity	k_r	0.5 W m ⁻¹ K ⁻¹	LITERATURE
Cell–cell conductance	G	0.02 W/K	ASSUMED
Resistance scatter	σ_R	5 %	ASSUMED
Duct + grille loss	K	3.5 + 1.4	ASSUMED
Contact + PCB heat	—	32.5 W	ASSUMED

4 Results

4.1 Why more fans do not help

At the one-sided operating point the bank contributes 1421.9 Pa of a 1454.9 Pa total — 97.7%. The fan is delivering 148 mmH₂O against a dead-head of 150 mmH₂O, which is to say it is operating on the flat, near-vertical part of its curve. Adding fans in parallel adds flow capacity where the curve has already collapsed.

4.2 Series staging

Pushing at one cutout while pulling at the other puts the two fan sets in series, so by (5) their pressure rises add. The system curve is unchanged; the available pressure is doubled, and the intersection moves.

Table 4: Fan count against outcome, both candidate flow axes. On the axis the real cutouts occupy, a four-fold increase in fan power is worth 0.10 °C.

Fans	flow $\parallel z$ (cutout axis)		flow $\parallel x$ (open axis)	
	Flow	Hottest cell	Flow	Hottest cell
1	78.6 CFM	48.21 °C	126.3 CFM	47.08 °C
3	93.2 CFM	47.02 °C	359.7 CFM	43.01 °C
6	94.4 CFM	46.94 °C	613.5 CFM	41.97 °C
12	94.7 CFM	46.92 °C	817.2 CFM	41.58 °C

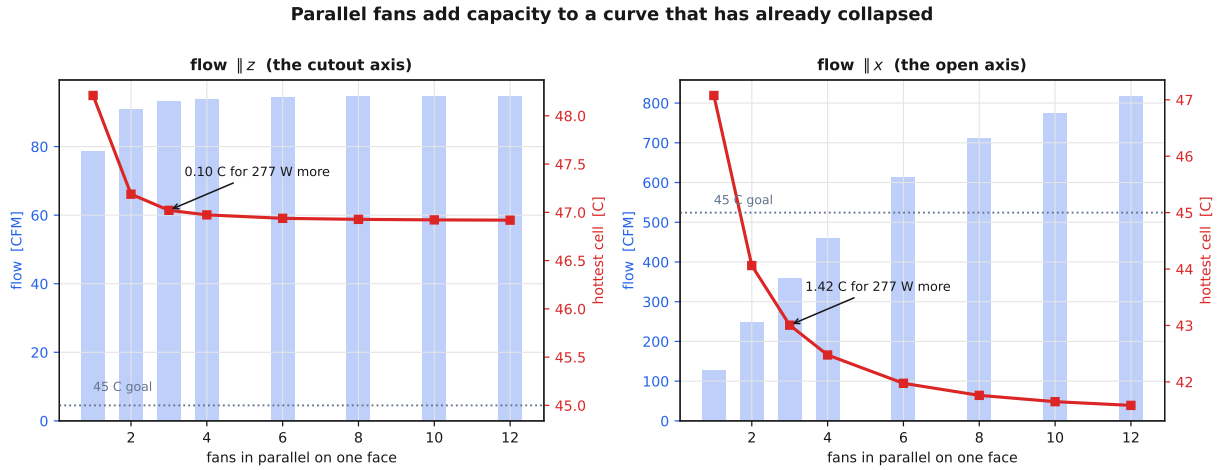


Figure 2: The same fans on the two candidate axes. On the cutout axis (left) the curve is flat in fan count; on the open axis (right) the same hardware more than doubles the flow. The difference is the free-area ratio: 0.102 against 0.349.

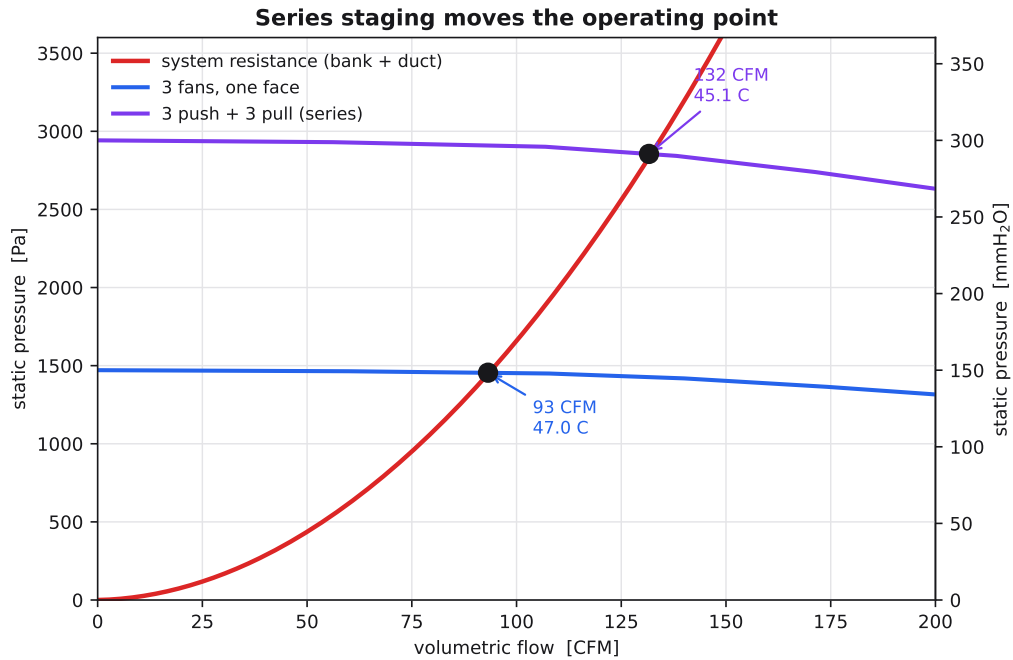


Figure 3: Fan characteristics against the system resistance. The red curve is the same in both cases — only the available pressure differs. Series staging moves the operating point from 93 CFM to 132 CFM.

Table 5: One-sided against push-pull at identical fan power.

Quantity	3 fans, one face	3 push + 3 pull
Flow	93.2 CFM	131.6 CFM
Face velocity	1.38 m/s	1.94 m/s
$V_{\max}$	13.43 m/s	18.96 m/s
$Re_{\max}$	14373	20391
h	$136.7 \text{ W m}^{-2}\text{K}^{-1}$	$168.3 \text{ W m}^{-2}\text{K}^{-1}$
Δp bank	1422 Pa	2789 Pa
Δp total	1455 Pa	2855 Pa
Air temperature rise	5.95 °C	4.21 °C
Exhaust air	45.95 °C	44.21 °C
Hottest cell	47.02 °C	45.12 °C
Fan power	75.6 W	151.2 W

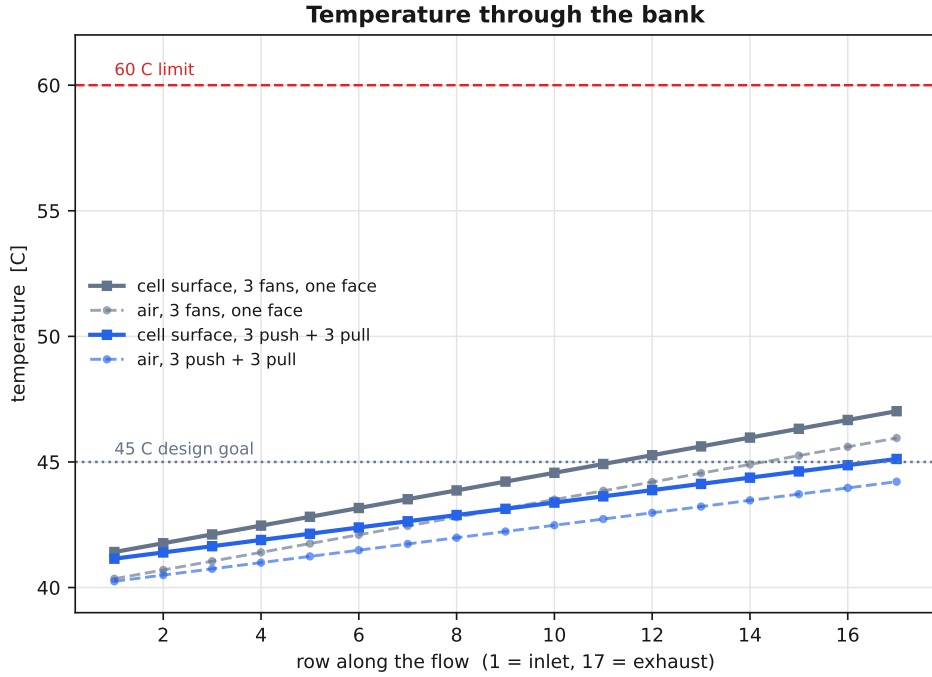


Figure 4: Air and cell temperature along the flow. Push-pull both lowers the whole profile and flattens it, because the larger mass flow cuts the air-side rise from 5.95 °C to 4.21 °C.

4.3 Placement search

All 30 physically realisable arrangements over the two cutouts were enumerated — each face either unused, blowing in, or drawing out, with up to three fans, subject to at least one path in and one out. Each costs a single root find, so exhaustive enumeration is both cheaper than an optimiser and provably optimal within the space.

4.4 Transient behaviour

The pack reaches 99 % of its final rise in roughly $5\tau \approx 200 \text{ s}$. At $\sigma_R = 5 \%$ the cell-to-cell spread is $-0.15\text{--}0.31 \text{ K}$ about each row mean — between 1 and 2 % of the 4.2 K row-to-row rise. *The scatter is therefore nearly invisible against the axial gradient*, which is worth stating plainly: a visualisation that makes it look dramatic is exaggerating. The tool exposes a separate gain control for that purpose, labelled as non-physical.

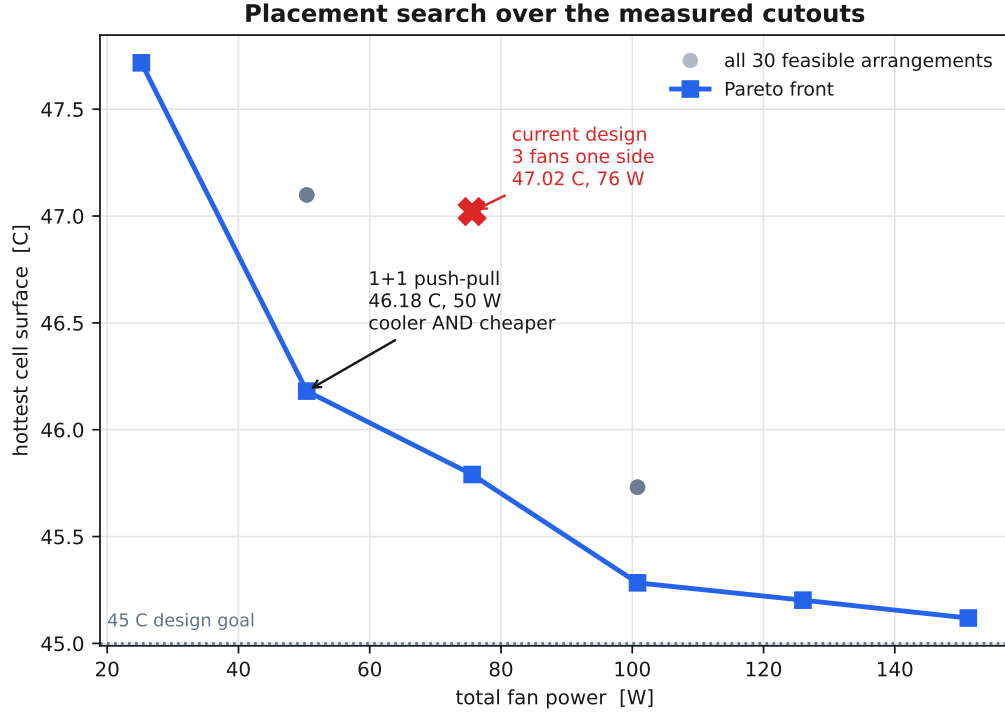


Figure 5: Peak cell temperature against fan power. The current design (red cross) is *dominated*: one fan on each face is both cooler and cheaper.

Table 6: Pareto front. Most of the available benefit arrives with the first fan on the second face, not with more fans on the first.

Arrangement	Flow	Hottest cell	Power
1 green in	84.1 CFM	47.72 °C	25.2 W
1 blue in + 1 green out	107.1 CFM	46.18 °C	50.4 W
1 blue in + 2 green out	115.0 CFM	45.79 °C	75.6 W
2 blue in + 2 green out	127.1 CFM	45.28 °C	100.8 W
2 blue in + 3 green out	129.3 CFM	45.20 °C	126.0 W
3 blue in + 3 green out	131.6 CFM	45.12 °C	151.2 W
<i>current: 3 one side</i>	93.2 CFM	47.02 °C	75.6 W

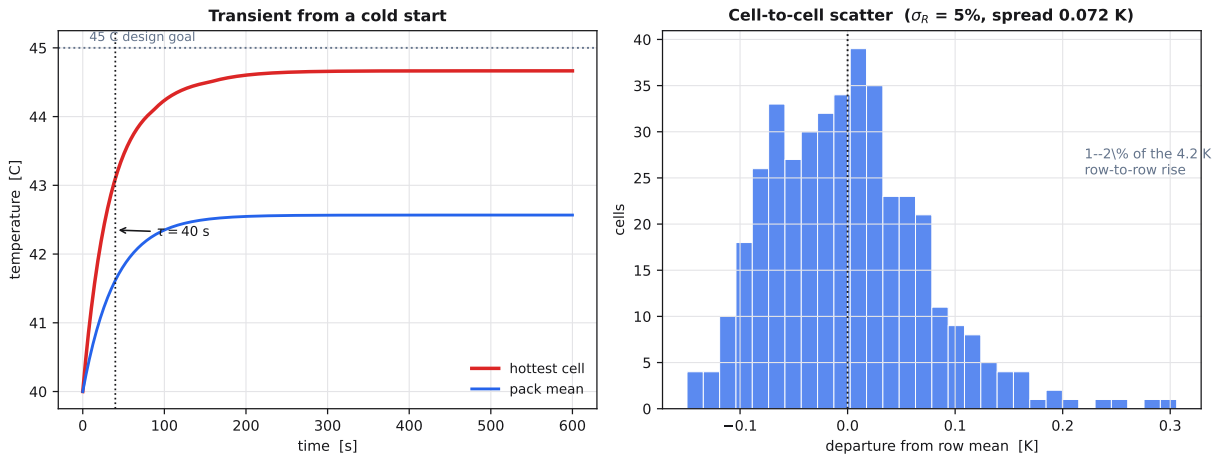


Figure 6: Left: approach to steady state from a cold start under push-pull, with the time constant $\tau = C/(hA) = 39.9\text{ s}$ marked. Right: the distribution of each cell's departure from its own row mean.

5 Verification

Table 7: Checks run against the implementation.

Check	Result
Network model reproduces the single-fan solver	exact, 0.0 CFM difference
Transient $t \rightarrow \infty$ against analytic steady state	2.8e-12 K
Exponential approach at $t = \tau$	0.6336 vs $1 - e^{-1} = 0.6321$
Transient energy balance	1.1e-15 relative
Steady energy balance	$< 1e - 3$ relative
$Re_{\max}$ within Žukauskas validity	14373–20391
Cells recovered from CAD	408
Both-inlet topology	raises, does not return a value

Against the design review’s own MATLAB study the model runs 11–22 % hotter across five published operating points. The bias is consistent and traceable to the friction extrapolation described below; it is conservative.

6 Limitations

These are stated in order of how much they matter.

1. **The friction extrapolation dominates everything.** The bank is 97.7 % of the system resistance, and its friction factor is extrapolated from the tightest published Žukauskas curve ($P_T = 1.25$) to the actual $P_T = 1.114$. Every pressure and flow in this report inherits that extrapolation. It over-predicts Δp and so under-predicts flow.
2. **The lumped-node assumption is formally violated.** With $Bi = h(D/4)/k_r = 1.55$, well above the usual 0.1 threshold, a node temperature is a *volume mean*, not a surface temperature. The centre-to-surface rise is 1.78 K — larger than the entire scatter effect — and is added explicitly before any comparison with the 60 °C surface limit.
3. **Several inputs are assumptions, not measurements.** The cell–cell conductance (0.02 W/K) is the weakest and the first worth measuring. Contactor and PCB dissipation (32.5 W) does not appear in the design review at all; because those bodies sit outside the cell air path it raises the exhaust by 0.46 K and leaves cell temperatures untouched.
4. **One-dimensional flow.** Uniform velocity across the face, no bypass around the bank, no recirculation, and heat spread evenly over the cells. Real packs leak around the edges, which would penalise the low-resistance configurations more than the high-resistance ones.
5. **The push–pull result is unvalidated.** No CFD or experimental data exists for it. The series composition assumes the two fan sets do not interfere aerodynamically — reasonable at 590 mm separation with the bank between them, but unverified.
6. **The flow direction is a modelling choice.** The fans and ducts were suppressed before the CAD export, so nothing in the geometry proves which way the air actually moves. The cutout dimensions make the z axis overwhelmingly likely, and that is what is modelled throughout.

7 Conclusions

- The pack passes the 60 °C limit in every configuration examined, with at least 11.8 °C of margin.

- The current three-fans-on-one-face arrangement reaches 47.02°C and misses the 45°C design goal by 2.02°C .
- **Moving one of those fans to the opposite cutout is cooler and cheaper** — 46.18°C at 50.4 W with one fan per face. Running three on each face reaches 45.12°C and effectively meets the goal.
- Adding fans to a single face is not a useful lever. The bank, not the fans, sets the flow.
- The largest single uncertainty is the friction extrapolation. If one measurement is to be made, a pressure-drop test across the bank at a known flow would retire most of the doubt in this report.